



BUSINESS BLUEPRINT

AOP AUTOMATION FOR C2 MEG PLANT

Honeywell

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Business Blueprint

AOP Automation for C2 MEG Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)



Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	12-Nov-2024	HON	Issued for Review
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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for C2 MEG plant for C2 site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various MEG plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for MEG technology.

1.2 Reference Documents

Document Name	Revision	Title	Document
C2 MEG DCT	-	Digital AOP C2 MEG DCT	 Digital AOP C2 MEG DCT.xlsx
UAT Observation list	-	Digital AOP MEG UAT Observation list	 Digital AOP MEG UAT Observations.xlsx
EOE effective Operating Hours calculation	-	RIL AOP - MEG Maintenance Screen - Slowdown Logic_R&T Validated	 RIL AOP - MEG Maintenance Screen
Slowdown Norms Methodology	-	Digital AOP - Normal_slowdown_shutdown_Sample Norms Calculation	 JMD_DCT_Updated_Portal Formats.xlsx

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Overall AOP Consumption

Steps to generate the Production and Overall AOP Consumption are captured in the below flow diagram.

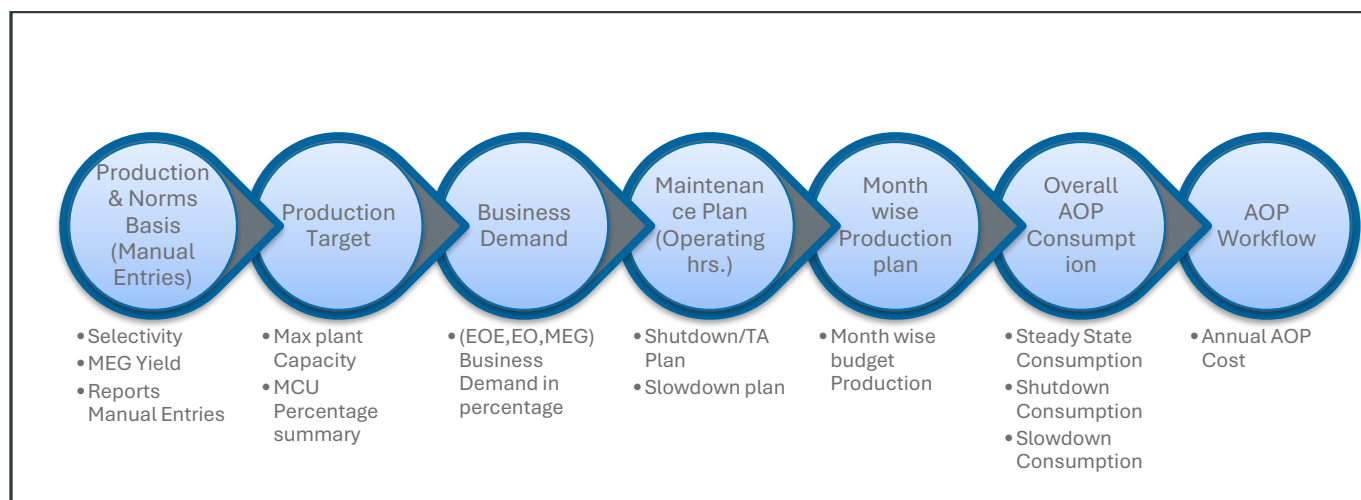


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Month wise Production plan

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production & Norms Basis

The production & norms basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data, including Yield, Selectivity, and Consumption, are configured under the production & norms basis screen to support Production and Overall AOP Consumption calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

The user must insert the TST value month wise and save it with a valid reason. This will provide the Latent heat against TST.

NOTE- Latent heat will be calculated only if the user provides the TST temperature; otherwise, the latent heat value will be set to 0.

In production & norms basis screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Report manual entry tab is provided to enter report specific manual entries for last budgeted FY actual values (e.g. Budgeted selectivity, Actual Selectivity, guaranteed selectivity and actual operating Hours)
- Option to import and export manually entered from production & norms basis screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.

Production & Norms Basis														
MEG / C2 / C2 MEG														
AOP Historical Period Basis														
Start Date 01-04-2020 End Date 31-03-2025 Load (Last refreshed data on: 20-05-2025 02:30 PM for the period from 01-04-2020 to 31-03-2025)														
AOP Design Basis 1. AOP design basis is considered for the period from 01-04-2020 to 31-03-2025.														
Configuration Constants Report Manual Entry (2024-25)														
Export Import Save														
Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	MT / MT	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272 HY ratio updat...
DEG Yield	MT / MT	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.101	0.103	0.101	0.102	0.102 DEG norms1
PEG Yield	MT / MT	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.022	0.021	0.006	0.006 PEG norms1
Calculation input for Norm														
Predicted Selectivity	%	88.15	88.13	88.03	87.97	87.91	87.85	87.79	87.73	87.58	87.86	87.8	87.82	Selectivity chan...
TST	C	246.4	246.56	246.8	247	247.2	247.4	247.6	247.8	248.6	248.35	248.55	248.25	TST
Total HHP Steam Consumption	TPH	64.5	65.5	65.5	65	65	65	65	65	64	64	64	65	HHPs monthwise
Total MP Steam Consumption	TPH	346	345	346	346	346	346	346	346	354	345	354	346	Total MP
Ethylene purity		1	1	1	1	1	1	1	1	1	1	1	1	1 from dct update
Oxygen purity	Mole%	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998 from dct update
MP Latent Heat	Kcal/kg	414.719	414.526	414.235	413.993	413.75	413.508	413.264	413.021	412.044	412.35	412.106	412.472	update2
HP Sparging	TPH	0	0	0	0	0	0	0	0	0	0	0	0	0 Not considered
Reactor Inlet Temp	°C	203	203.16	203.4	203.6	203.8	204	204.2	204.4	205.2	204.95	205.15	204.85	Click to add re...

Production & Norms Basis

Year: 2025-26

Vertical: MEG

Site: C2

Plant: C2 MEG

1

2

3

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11

Production & Norms Basis

Shutdown/TA Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

Overall AOP Consumption (Norm/Quantity)

MEG / C2 / C2 MEG

AOP Historical Period Basis

Start Date01-04-2020

End Date31-03-2025

Load

(Last refreshed data on: 20-05-2025 02:30 PM for the period from 01-04-2020 to 31-03-2025)

AOP Design Basis

1. AOP design basis is considered for the period from 01-04-2020 to 31-03-2025.

Configuration

Constants

Report Manual Entry (2024-25)

Export

Import

Save

Particulars	UOM	Value	Remark
Constant			
BFW Feed Temperature	°C	180.0	From DCT
Blowdown rate	TPH	10.0	From DCT
Cycle gas CP	kcal/kg/°C	0.545	CG cp
Cycle gas flow	kg/h	1970000.0	Actual
EO molecular weight		44.05	Actual
Ethylene concentration in argon purge	mol%	0.327	Actual
Ethylene Molecular Weight		28.05	Actual
Methane concentration in argon purge	Mol %	0.525	From DCT
Methane molecular weight		16.0	From DCT
Oxygen molecular weight		32.0	From DCT
Water Cp	kcal/kg/°C	1.14	From DCT
Oxygen concentration in argon purge	mol%	0.065	From DCT
Argon purge Molecular weight		22.0	From DCT
Steam Drum BFW Blowdown rate	TPH	10.0	From DCT
BFW Sensible Heating delta T	°C	5.0	From DCT
Additional Condensate for Condensate Return		59.0	From DCT

Production & Norms Basis														
Report Manual Entry (2024-25)														
Export Import Save														
Particulars	UOM	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Remark
Report Manual Entry														
Budgeted Selectivity	%	88.15	88.06	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4	Selectivity drop ...
Actual Selectivity	%	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4	TEST
Guaranteed Selectivity	%	86.2	85.9	85.6	85.3	85	84.7	84.3	84	83.9	83.6	83.4	83.2	TEST
Actual Operating Hours	Hrs	720	744	720	744	744	720	744	720	744	744	672	744	test3

Figure 2 Digitized AOP screen for Production & Norms Basis

2.1.1.2 Production Target

The Max Capacity (MC) of MEG plants is defined on an EOE basis.

MC values for Ethylene Oxide Equivalent (EOE) and Monoethylene Glycol (MEG) will be captured separately.

Month wise data is considered for deriving the best value. This will account for seasonal effects as well as the festive season.

If AOP basis is considered as 05 years, value estimation involves averaging production data over continuous 5-day periods in the last 5 years, after removing outliers and considering the normal operating mode without recycle for each month, 5 such averages are calculated and the maximum average value from the available 05 average set is designated as the MC value or production volume. Also, the edit option provided to change the derived MC value.

NOTE- AOP historical period basis added in production & norms basis screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive month-wise EOE MC values.

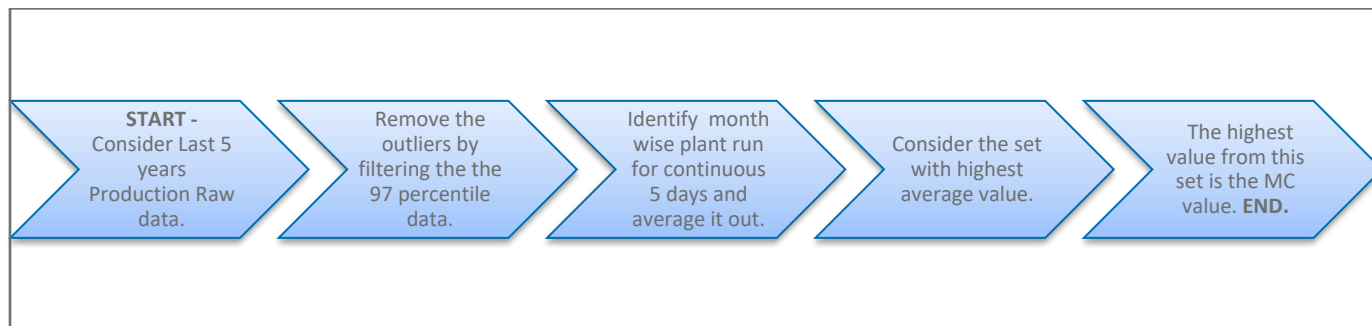


Figure 3 Digitized AOP Steps to derive EOE month wise MC values

The MEG MC value is determined using the following formula.

$$MEG\ MC = MEG\ Yield * EOE\ MC$$

Where MEG yield is a month-wise manual entry captured in the AOP digitization portal in the Production & norms basis screen, as seen in the screenshot below.

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	MT / MT	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272 HY ratio updati...
DEG Yield	MT / MT	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.101	0.103	0.101	0.102 DEG norms1
PEG Yield	MT / MT	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.022	0.021	0.006 PEG norms1

Figure 4 Digitized AOP screen for MEG Yield

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary, comparing each month's value against the highest MU value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.



Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	94	94	94	94 MCU normal + High load 2
MEG	116.706	116.706	116.706	116.706	116.706	116.706	116.706	116.706	116.706	119.568	119.568	119.568	119.568 Click to add remark

Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	97.61	97.61	97.61	97.61	97.61	97.61	97.61	97.61	97.61	100	100	100	100
MEG	97.61	97.61	97.61	97.61	97.61	97.61	97.61	97.61	97.61	100	100	100	100

Figure 5 Digitized AOP screen for Production Target

The '**Calculate**' button on this screen allows the user to compute the MEG MC value if any changes have been made to the MEG Yield and Start and End date for AOP basis selection on the production & norms basis screen.

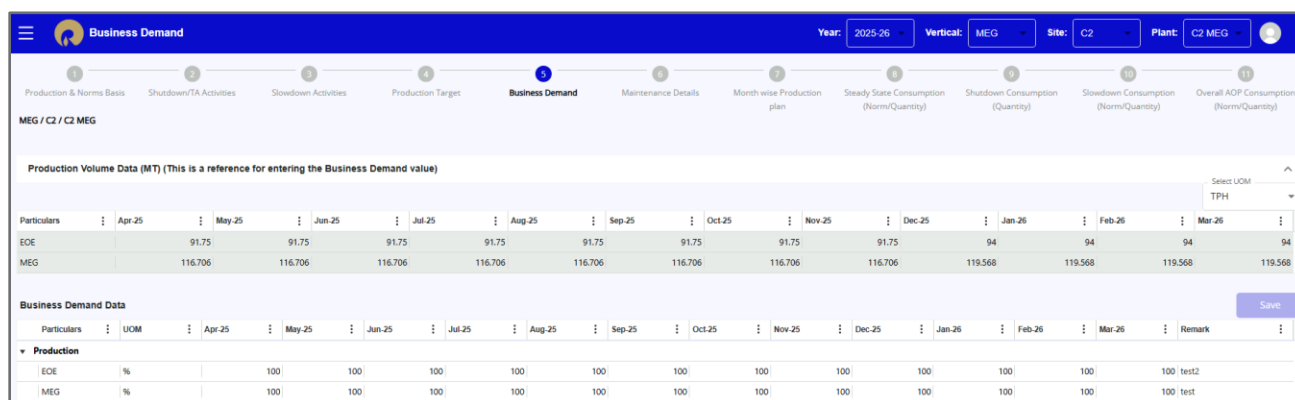
2.1.1.3 Business Demand

The production demand is considered as the maximum value from last year's achieved production values. The default value for EOE and MEG is 100%.

The business demand value can be modified by users upon providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

NOTE- For the C2 MEG plant, CO2 demand will be captured in this section.



Particulars	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
EOE	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	94	94	94
MEG	116.706	116.706	116.706	116.706	116.706	116.706	116.706	116.706	116.706	119.568	119.568	119.568

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
EOE	%	100	100	100	100	100	100	100	100	100	100	100	100	test2
MEG	%	100	100	100	100	100	100	100	100	100	100	100	100	test

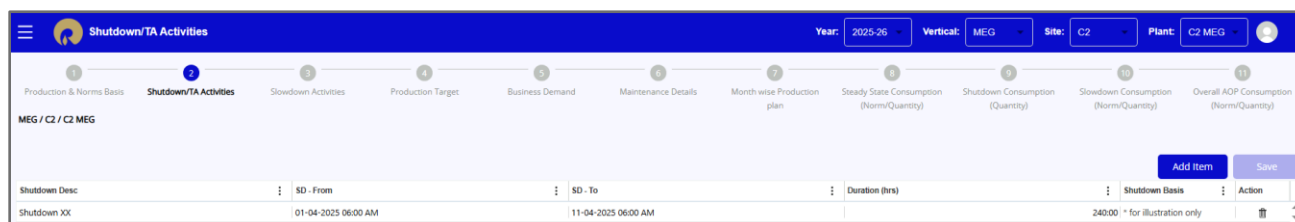
Figure 6 Digitized AOP screen for Business Demand

2.1.1.4 Maintenance/Shutdown/Slowdown/TA Activities

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The below screenshot depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.



Shutdown Desc	SD - From	SD - To	Duration (hrs)	Shutdown Basis	Action
Shutdown XI	01-04-2025 06:00 AM	11-04-2025 06:00 AM		240.00 * for illustration only	

Figure 7 Digitized AOP screen for Shutdown/Turnaround Activities

The Slowdown entries will have the inputs as shutdowns, along with the requirement of plant EOE load (TPH basis) for the duration of the slowdown.

The slowdown activities can be calculated based on the slowdown rate and hours. This is called a Basis Shutdown, equivalent to Shutdown.

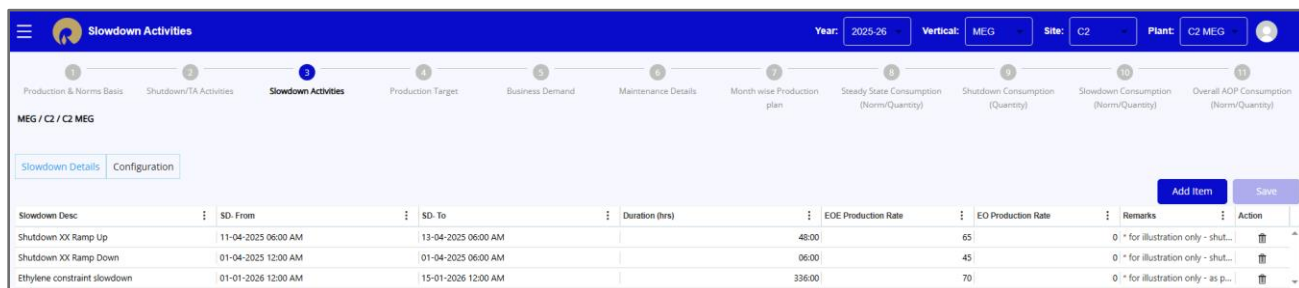
$$\text{Slowdown rate(TPH)} = (\text{MCU} - \text{Operating Production rate})$$

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate(TPH)} * \text{Slowdown hrs}) / \text{MCU}$$

Operating Production rate will be a manual entry in Slowdown activity screen and slowdown rate will be calculate at the backend as per above formula.

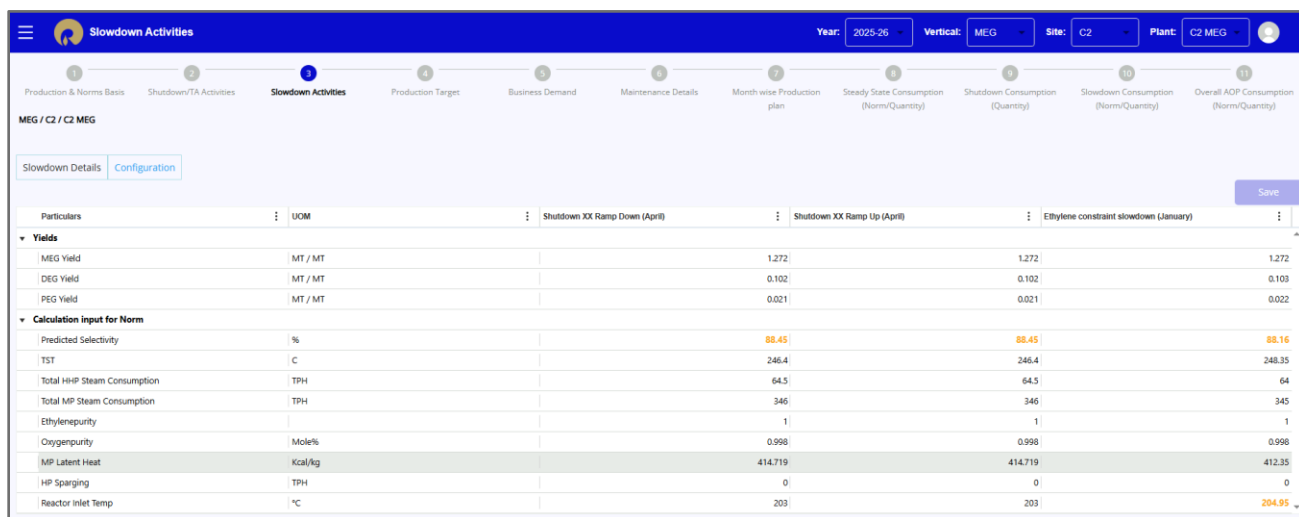
For each shutdown entry, two slowdown entries will be generated automatically, with the

nomenclature as “#Shutdown Entry name. – Ramp down” and “#Shutdown Entry name. – Ramp Up”. In Slowdown details tab, EOE production rate (TPH) and EO production rate (TPH) these two columns will be added in Slowdown activities screen as shown below.



Shutdown Desc	SD- From	SD- To	Duration (hrs)	EOE Production Rate	EO Production Rate	Remarks	Action
Shutdown XX Ramp Up	11-04-2025 06:00 AM	13-04-2025 06:00 AM		48.00	65	0 * for illustration only - shut...	
Shutdown XX Ramp Down	01-04-2025 12:00 AM	01-04-2025 06:00 AM		06.00	45	0 * for illustration only - shut...	
Ethylene constraint slowdown	01-01-2026 12:00 AM	15-01-2026 12:00 AM		336.00	70	0 * for illustration only - as p...	

Separate tab (Configuration) will be provided as shown below in slowdown activity screen for the provision of manual entry related to production & norms basis screen parameters based on the type of slowdown. Number of rows mentioned in slowdown activity screen will be no of column for slow down manual entry screen.



Particulars	UOM	Shutdown XX Ramp Down (April)	Shutdown XX Ramp Up (April)	Ethylene constraint slowdown (January)
Yields				
MEG Yield	MT / MT		1.272	1.272
DEG Yield	MT / MT		0.102	0.102
PEG Yield	MT / MT		0.021	0.021
Calculation input for Norm				
Predicted Selectivity	%		88.45	88.45
TST	C		246.4	246.4
Total HHP Steam Consumption	TPH		64.5	64.5
Total MP Steam Consumption	TPH		346	346
Ethylene purity			1	1
Oxygen purity	Mole%		0.998	0.998
MP Latent Heat	Kcal/kg		414.719	414.719
HP Sparging	TPH		0	0
Reactor Inlet Temp	°C		203	203

Figure 8 Digitized AOP screen for Slowdown Activities

The 'Add Item' button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

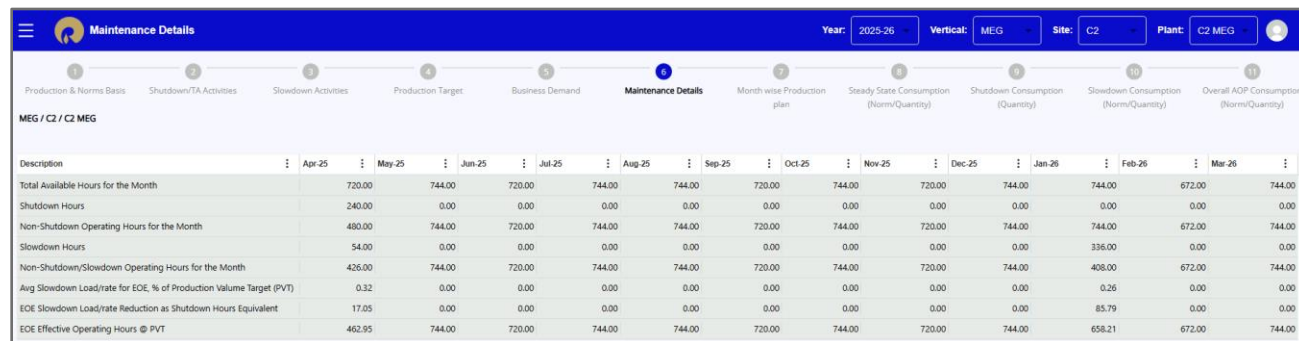
For each month, the following table is generated based on all inputs for shutdowns/slowdowns-

Description	April	...	March
Total Available Hours for the Month	$30 \times 24 = 720$		
Shutdowns Hours	<i>From the Routine Shutdown Plan Section</i>		
Non-Shutdown Operating Hours for the Month	<i>Total Available operating hours for the month – Shutdown hrs</i>		
Slowdown Hours	<i>From the Slowdown activities details</i>		
Non-Shutdown/Slowdown Operating Hours for the Month	Non Shutdown Operating Hours for the Month – Slowdown Hours		
Average Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i \times EOE \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i) \times EOE \text{ MC for Month}}$		
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EOE \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EOE \text{ slowdown hrs}_i)$		
EOE Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EOE Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		

Table 3 Maintenance Details

The screen below depicts the Maintenance Details Table, which is captured in the digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.



Description	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Shutdown Hours	240.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Non-Shutdown Operating Hours for the Month	480.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Slowdown Hours	54.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	336.00	0.00	0.00
Non-Shutdown/Slowdown Operating Hours for the Month	426.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	408.00	672.00	744.00
Avg Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	17.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	85.79	0.00	0.00
EOE Effective Operating Hours @ PVT	462.95	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	658.21	672.00	744.00

Figure 9 AOP Maintenance Details Table

2.1.1.5 Calendar Year Based Production Calculations Details/Month wise Production plan

Using the Business Demand, Production Target and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\text{Production AOP (MT)} = (\text{Operating hours for EOE/EO in a month (hrs)} * \text{Percentage of product demand} * \text{MC Value of product(TPH)}) / 100$$

$$\text{Operating hours for MEG} = \text{Total Available hours of the month} - \text{Shutdown/TA hours of the month}$$

$$\text{Operating hours for EOE} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

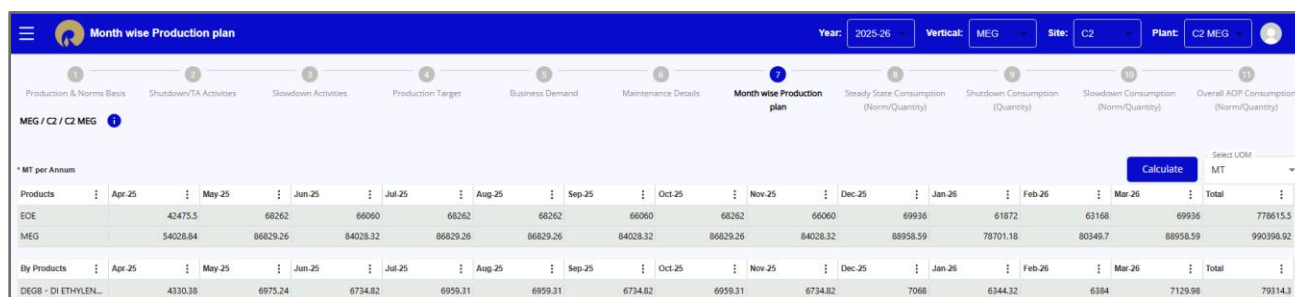
$$\text{Operating hours for EO} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

Hence, month-wise production is,

$$\text{Total Budgeted Production} = \sum \text{All the values for the month}$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.



Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
EOE	42475.5	68262	66060	68262	68262	66060	68262	66060	69936	61872	63168	69936	778615.5
MEG	54028.84	86829.26	84028.32	86829.26	86829.26	84028.32	86829.26	84028.32	88958.59	78701.18	80349.7	88958.59	990398.92

By Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
DEGB - DI ETHYLENE	4330.38	6975.24	6734.82	6959.31	6959.31	6734.82	6959.31	6734.82	7068	6344.32	6384	7129.98	79314.3

Figure 10 Digitized AOP screen for Month wise Production plan

The 'Calculate' button on this screen allows users to recalculate Production values when changes are made to production & norms basis settings (e.g., MEG yield), business demand, or when line items are added to slowdown and shutdown activities.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Consumption

All the specific Overall AOP Consumption are set with One-unit EOE as the basis.

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

Reliance Industries Ltd
AOP Automation for C2 MEG Plant

Proprietary and Confidential
Document No.
September 18, 2025

1. Historical Best
2. Based on Calculations
3. Manual Entries

3.1.2.1.1 Historical Best

Based on the historical data of last 5 years, corresponding to the MCU production, the best-achieved feed slate period is filtered. This data is further used for the calculation of consumption norms, which is the average of the norms during that period, excluding the identified outliers.

Below is the logic formulated to derive the Historical best Normal Operation Norms –

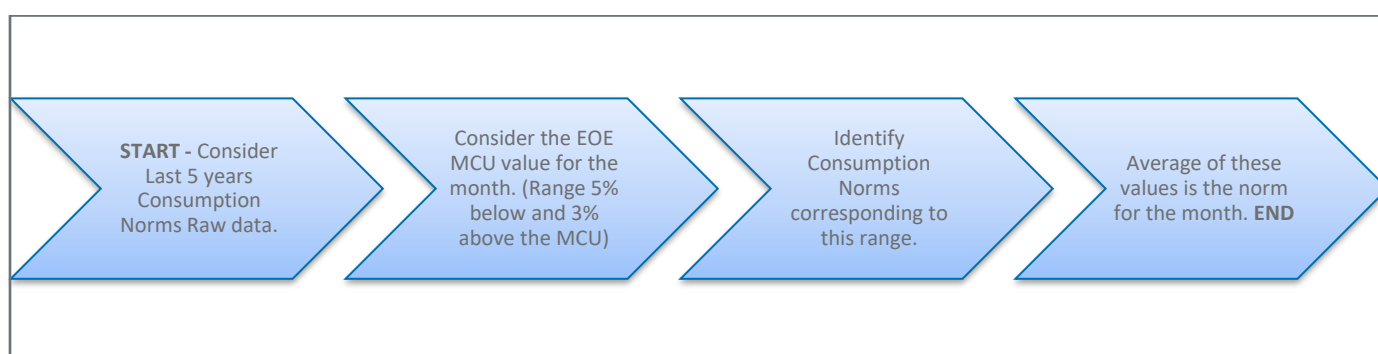


Figure 11 Digitized AOP Steps to derive month wise Historical Best Norms

The user has provision to overwrite/update the value by providing a valid reason for the historical best particulars.

The list of particulars to be derived using the Historical Best logic is specified in the DCT provided by the user. These particulars may vary from plant to plant.

3.1.2.1.2 Based on Calculations

For the MEG vertical, particulars like ethylene, oxygen, HP Steam, BFW water etc. are formulated based on the stoichiometric calculations provided by the user in the DCT.

The formula can vary from plant to plant.

E.g. the Raw Material norms for ethylene and oxygen for C2 MEG are derived using the calculations mentioned in the below table.

Plant	Category	Particular	Calculation
C2 MEG	Raw Material	Ethylene	$(28.05/44.05 \times 100 / \{selectivity\} / 0.9995) + ([METHANE] \times 28.05 \times 0.32 / 16 / 0.525) / 0.9995$ Where, 28.05 - Ethylene Molecular Weight 44.05 - EO molecular weight 0.9995 - Ethylene Purity (wt%) 0.32 - Ethylene concentration in argon purge (Mol %) 16 - Methane molecular weight 0.525 - Methane concentration in argon purge (Mol %)
		Oxygen	$((100 \times 2.1818 / \{selectivity\} - 1.818) + ([METHANE] \times 32 \times 0.065 / 16 / 0.525)) / 0.998$ Where, 2.1818, 1.881 - Stoichiometric Ratios 44.05 - EO molecular weight 32 - Oxygen molecular weight 0.065 - Oxygen concentration in argon purge (Mol %) 16 - Methane molecular weight 0.525 - Methane concentration in argon purge (Mol %) 0.998 - Oxygen Purity (wt%)

Table 4 Raw Material Norms calculations

The values for *losses and purity* can be captured from the Lab analysis or Past/last year's operations. The user has the provision to enter the value manually in the Production & norms basis screen.

Selectivity, Losses and Purity can be modified/added by the user in the Production & norms basis Screen under the Calculation Input for Norms Subsection.

Production & Norms Basis

Year:

2025-26

Vertical:

MEG

Site:

C2

Plant:

C2 MEG

1

Production & Norms Basis

2

Shutdown/TA Activities

3

Slowdown Activities

4

Production Target

5

Business Demand

6

Maintenance Details

7

Month wise Production plan

8

Steady State Consumption (Norm/Quantity)

9

Shutdown Consumption (Quantity)

10

Slowdown Consumption (Norm/Quantity)

11

Overall AOP Consumption (Norm/Quantity)

MEG / C2 / C2 MEG

AOP Historical Period Basis

Start Date

01-04-2020

End Date

31-03-2025

Load

(Last refreshed data on: 20-05-2025 02:30 PM for the period from 01-04-2020 to 31-03-2025)

AOP Design Basis

1. AOP design basis is considered for the period from 01-04-2020 to 31-03-2025.

Configuration

Constants

Report Manual Entry (2024-25)

Export

Import

Save

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	MT / MT	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272	1.272: HY ratio updati...
DEG Yield	MT / MT	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.101	0.103	0.101	0.102	DEG norms1
PEG Yield	MT / MT	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.022	0.021	0.006	PEG norms1
Calculation input for Norm														
Predicted Selectivity	%	88.15	88.13	88.03	87.97	87.91	87.85	87.79	87.73	87.58	87.86	87.8	87.82	Selectivity chan...
TST	C	246.4	246.56	246.8	247	247.2	247.4	247.6	247.8	248.6	248.35	248.55	248.25	TST
Total HHP Steam Consumption	TPH	64.5	65.5	65.5	65	65	65	65	65	64	64	64	65	HHPs monthwise
Total MP Steam Consumption	TPH	346	345	346	346	346	346	346	346	354	345	354	346	Total MPS
Ethylene purity		1	1	1	1	1	1	1	1	1	1	1	1	1: from dct update
Oxygen purity	Mole%	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998: from dct update
MP Latent Heat	Kcal/kg	414.719	414.526	414.235	413.993	413.75	413.508	413.264	413.021	412.044	412.35	412.106	412.472	update2
HP Sparging	TPH	0	0	0	0	0	0	0	0	0	0	0	0	0: Not considered
Reactor Inlet Temp	°C	203	203.16	203.4	203.6	203.8	204	204.2	204.4	205.2	204.95	205.15	204.85	Click to add re...

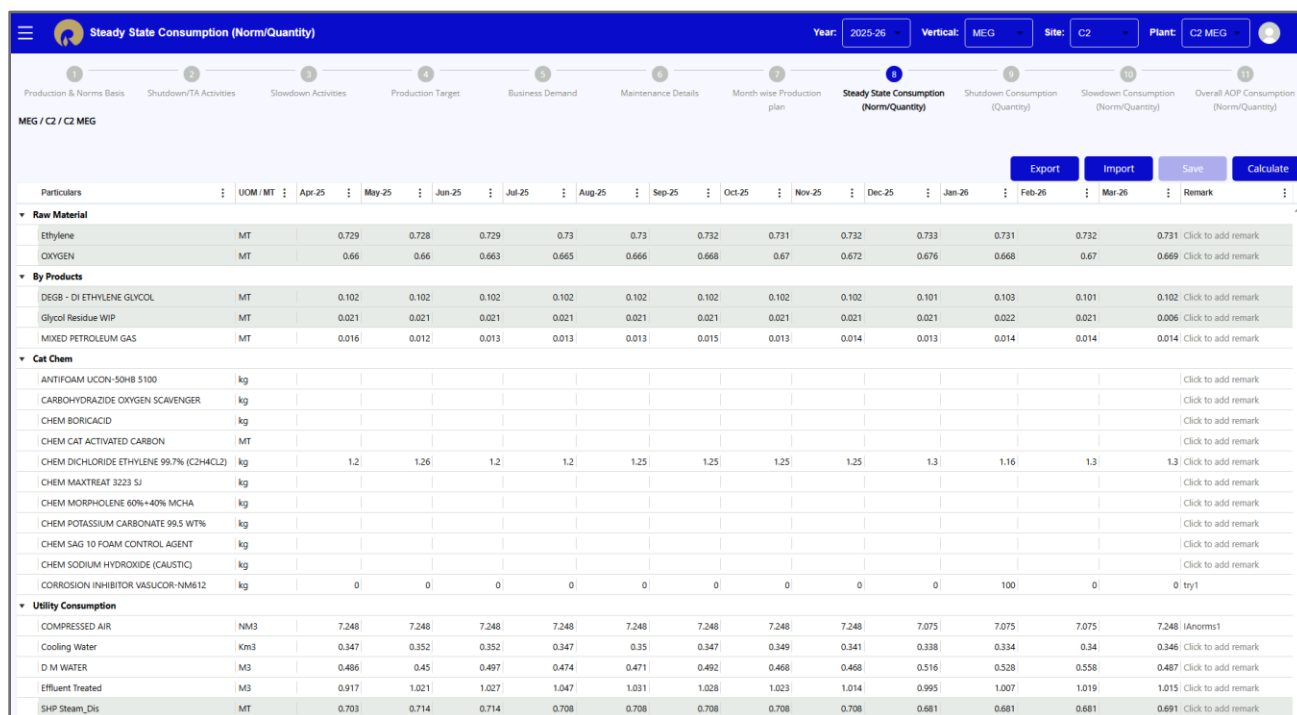
Figure 12 Digitized AOP screen for Catalyst Selectivity month wise

In the Steady State Consumption screen the cells for particulars with formula, e.g. HP Steam Dis, will be read only. If the values for such particulars are to be modified then the respective inputs should be modified, e.g. Total HP Steam consumption in the Production & norms basis Screen.

3.1.2.1.3 Manual Entries

The particulars under the Chemicals and Catalyst sections are manually entered by the user, who must also provide the corresponding norm values in the Normal Operation Norms screen.

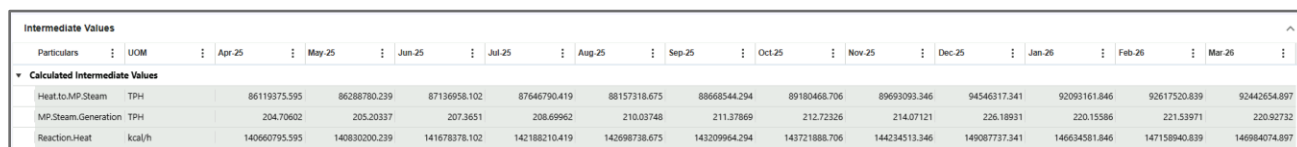
The screenshot below highlights the details captured in the digitized AOP platform in the Steady State Consumption screen.



Particulars	UOM / MT	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT	0.729	0.728	0.729	0.73	0.73	0.732	0.731	0.732	0.733	0.731	0.732	0.731	Click to add remark
OXYGEN	MT	0.66	0.66	0.663	0.665	0.666	0.668	0.67	0.672	0.676	0.668	0.67	0.669	Click to add remark
By Products														
DEGB - DI ETHYLENE GLYCOL	MT	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.102	0.101	0.103	0.101	0.102	Click to add remark
Glycol Residue WIP	MT	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.022	0.021	0.006	Click to add remark
MIXED PETROLEUM GAS	MT	0.016	0.012	0.013	0.013	0.013	0.015	0.013	0.014	0.013	0.014	0.014	0.014	Click to add remark
Cat Chem														
ANTIFOAM UCON-50HB 5100	kg													Click to add remark
CARBOHYDRAZIDE OXYGEN SCAVENGER	kg													Click to add remark
CHEM BORICACID	kg													Click to add remark
CHEM CAT ACTIVATED CARBON	MT													Click to add remark
CHEM DICHLORIDE ETHYLENE 99.7% (C2H4CL2)	kg	1.2	1.26	1.2	1.2	1.25	1.25	1.25	1.25	1.3	1.16	1.3	1.3	Click to add remark
CHEM MAXTREAT 3223 SJ	kg													Click to add remark
CHEM MORPHOLINE 60%+40% MOHA	kg													Click to add remark
CHEM POTASSIUM CARBONATE 99.5 WT%	kg													Click to add remark
CHEM SAG 10 FOAM CONTROL AGENT	kg													Click to add remark
CHEM SODIUM HYDROXIDE (CAUSTIC)	kg													Click to add remark
CORROSION INHIBITOR VALUCOR-NM612	kg	0	0	0	0	0	0	0	0	0	100	0	0	try1
Utility Consumption														
COMPRESSED AIR	NM3	7.248	7.248	7.248	7.248	7.248	7.248	7.248	7.248	7.075	7.075	7.075	7.248	1Anorms1
Cooling Water	Km3	0.347	0.352	0.352	0.347	0.35	0.347	0.349	0.341	0.338	0.334	0.34	0.345	Click to add remark
D M WATER	M3	0.486	0.45	0.497	0.474	0.471	0.492	0.468	0.468	0.516	0.528	0.558	0.487	Click to add remark
Effluent Treated	M3	0.917	1.021	1.027	1.047	1.031	1.028	1.023	1.014	0.995	1.007	1.019	1.015	Click to add remark
SHP Steam_Dis	MT	0.703	0.714	0.714	0.708	0.708	0.708	0.708	0.708	0.681	0.681	0.681	0.691	Click to add remark

Figure 13 Digitized AOP screen for Steady State Consumption

The Normal Operation Normas screen also captures the Intermediate calculations required for Norms calculation as shown in the below screen shot –



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Calculated Intermediate Values													
Heat.to.MP.Steam	TPH	86119375.595	86288780.239	87136958.102	87646790.419	88157318.675	88668544.294	89180468.706	89693093.346	94546317.341	92093161.846	92617520.839	92442654.897
MP.Steam.Generation	TPH	204.70602	205.20337	207.3651	208.69962	210.03748	211.37869	212.72326	214.07121	226.18931	220.15586	221.53971	220.92732
Reaction.Heat	kcal/h	140660795.595	140830200.239	141678378.102	142188210.419	142698738.675	143209964.294	143721888.706	144234513.346	149087737.341	146634581.846	147158940.839	146984074.897

Figure 14 Digitized AOP screen for Steady State Consumption (Calculated Intermediate values)

The **'Calculate'** button on the screen will enable if the user makes any change in the dependent inputs in the Production & norms basis screen.

The **'Save'** button on the above screen enables the user to save any norm value updated manually

for historical best norms for cat chem.

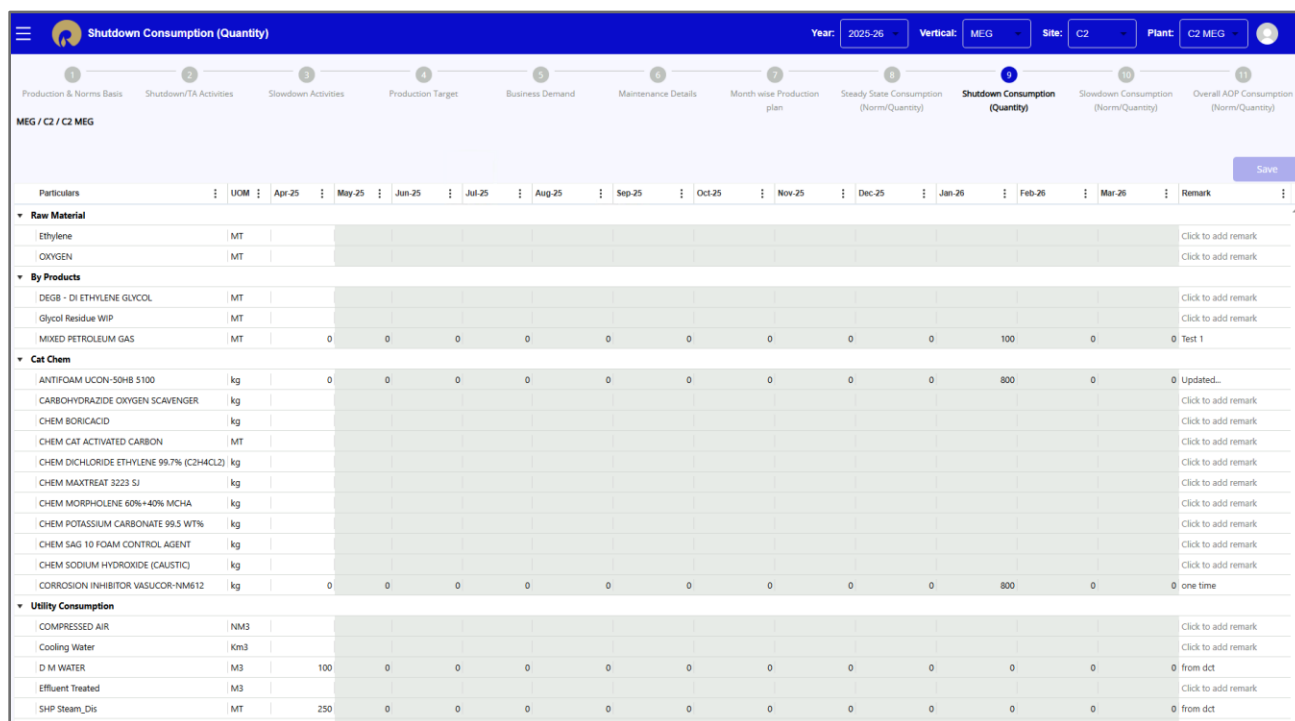
2.1.2.2 Shutdown Consumption

The Shutdown Consumption are recorded for the months in which shutdown activities are planned, as specified in the Shutdown Activities screen.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc..

The user must enter the consumption data (absolute values) for each line item.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT													Click to add remark
OXYGEN	MT													Click to add remark
By Products														
DEGB - DI ETHYLENE GLYCOL	MT													Click to add remark
Glycol Residue WIP	MT													Click to add remark
MIXED PETROLEUM GAS	MT	0	0	0	0	0	0	0	0	0	100	0	0	Test 1
Cat Chem														
ANTI-FOAM UCOR-50H 5100	kg	0	0	0	0	0	0	0	0	0	800	0	0	Updated...
CARBOHYDRAZIDE OXYGEN SCAVENGER	kg													Click to add remark
CHEM BORICACID	kg													Click to add remark
CHEM CAT ACTIVATED CARBON	MT													Click to add remark
CHEM DICHLORIDE ETHYLENE 99.7% (C2H4CL2)	kg													Click to add remark
CHEM MAXTREAT 3223 SJ	kg													Click to add remark
CHEM MORPHOLINE 60%+40% MCHA	kg													Click to add remark
CHEM POTASSIUM CARBONATE 99.5 WT%	kg													Click to add remark
CHEM SAG 10 FOAM CONTROL AGENT	kg													Click to add remark
CHEM SODIUM HYDROXIDE (CAUSTIC)	kg													Click to add remark
CORROSION INHIBITOR VASUCOR-NM612	kg	0	0	0	0	0	0	0	0	0	800	0	0	one time
Utility Consumption														
COMPRESSED AIR	NM3													Click to add remark
Cooling Water	Km3													Click to add remark
D M WATER	M3	100	0	0	0	0	0	0	0	0	0	0	0	from dct
Effluent Treated	M3													Click to add remark
SHP Steam_Dis	MT	250	0	0	0	0	0	0	0	0	0	0	0	from dct

Figure 14 Digitized AOP screen for Shutdown Consumption

2.1.2.3 Slowdown Consumption

Slowdown Consumption are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

Slowdown Consumption (Norm/Quantity)

Year: 2025-26

Vertical: MEG

Site: C2

Plant: C2 MEG

1

2

3

4

5

6

7

8

9

10

11

Production & Norms Basis

Shutdown/T/A Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

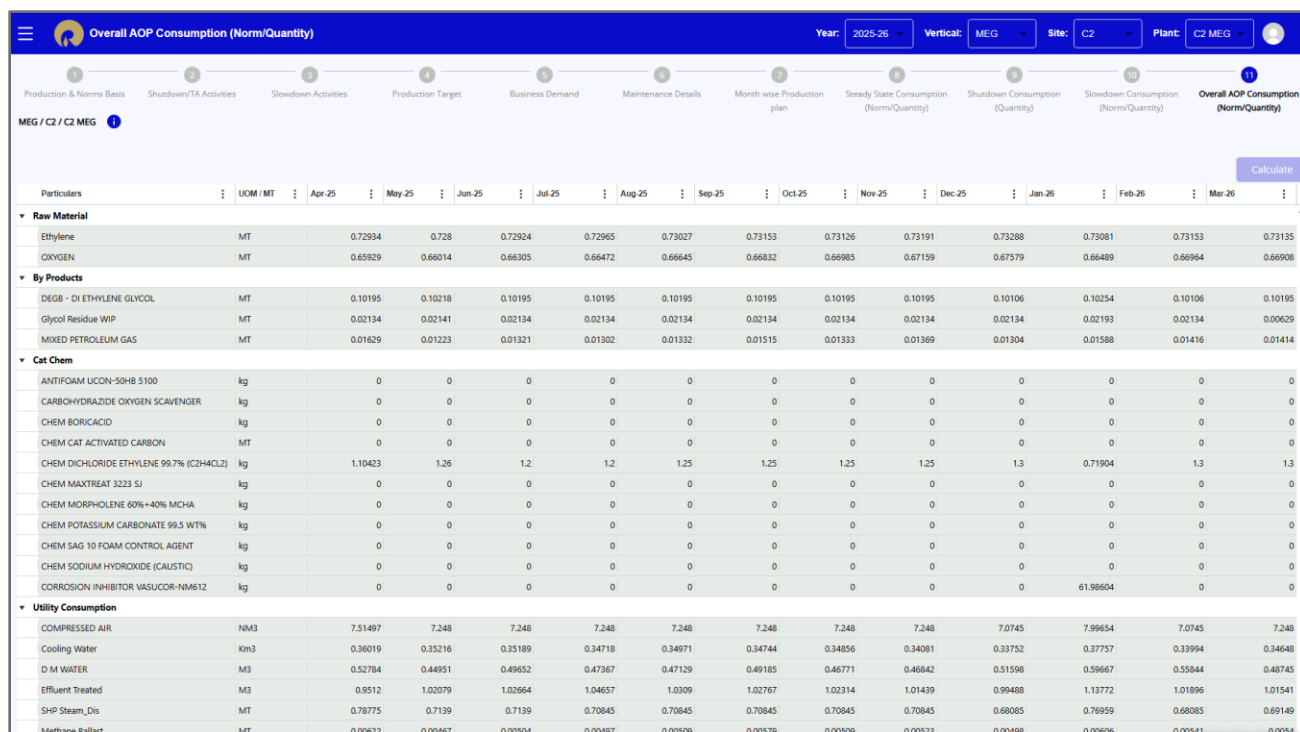
Overall AOP Consumption (Norm/Quantity)

MEG / C2 / C2 MEG

Save

Calculate

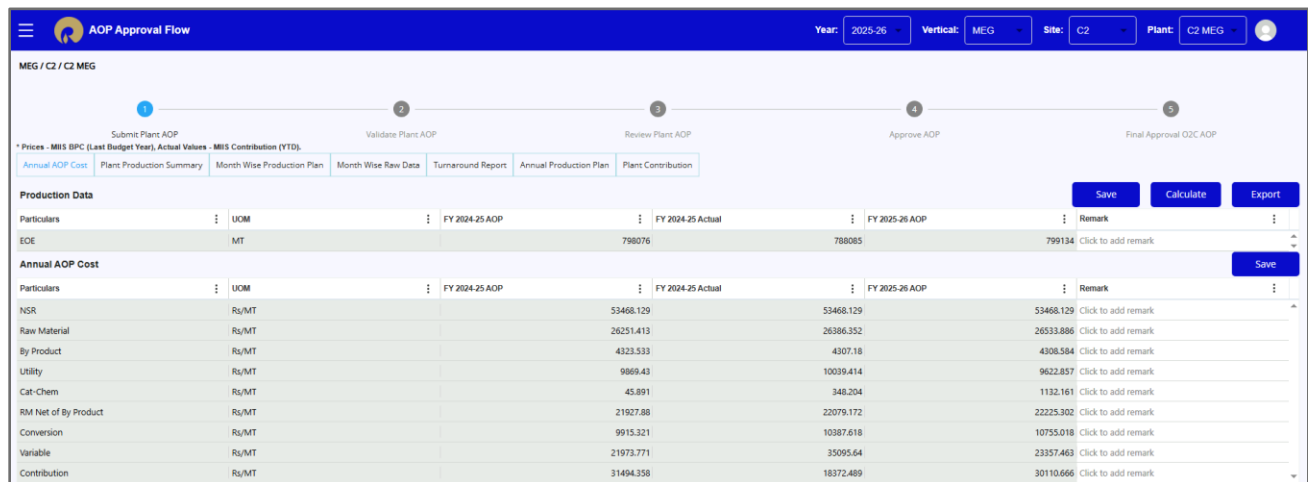
Particulars	UOM	Shutdown XX Ramp Down (April)	Shutdown XX Ramp Up (April)	Ethylene constraint slowdown (January)	
▼ Raw Material					
Ethylene	MT		0.734	0.73	0.731
OXYGEN	MT		0.653	0.652	0.66
▼ By Products					
DEGB - DI ETHYLENE GLYCOL	MT		0.102	0.102	0.103
Glycol Residue WIP	MT		0.021	0.021	0.022
MIXED PETROLEUM GAS	MT		0.032	0.022	0.019
▼ Cat Chem					
ANTIFOAM UCON-50HB 5100	kg				
CARBOHYDRIZADE OXYGEN SCAVENGER	kg				
CHEM BORICACID	kg				
CHEM CAT ACTIVATED CARBON	MT				
CHEM DICHLORIDE ETHYLENE 99.7% (C2H4CL2)	kg				
CHEM MAXTREAT 3223 SJ	kg				
CHEM MORPHOLENE 60%+40% MCHA	kg				
CHEM POTASSIUM CARBONATE 99.5 WT%	kg				
CHEM SAG 10 FOAM CONTROL AGENT	kg				
CHEM SODIUM HYDROXIDE (CAUSTIC)	kg				
CORROSION INHIBITOR VASUCOR-NM612	kg				
▼ Utility Consumption					
COMPRESSED AIR	NM3		14.778	10.231	9.5
Cooling Water	Km3		0.708	0.49	0.449
D M WATER	M3		0.992	0.687	0.709
Effluent Treated	M3		1.87	1.295	1.352
SHP Steam_Dis	MT		1.433	0.992	0.914



3. AOP Reports

- Dataset (Basis)
- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).
- Plant Contribution Summary (T-22).

The below screenshot depicts the AOP report captured in the AOP digitized platform.



Particulars	UOM	FY 2024-25 AOP	FY 2024-25 Actual	FY 2025-26 AOP	Remark
NSR	Ru/MT	53468.129	53468.129	53468.129	Click to add remark
Raw Material	Ru/MT	26251.413	26386.352	26533.886	Click to add remark
By Product	Ru/MT	4323.533	4307.18	4308.584	Click to add remark
Utility	Ru/MT	9869.43	10039.414	9622.857	Click to add remark
Cat-Chem	Ru/MT	45.891	348.204	1132.161	Click to add remark
RM Net of By Product	Ru/MT	21927.88	22079.172	22225.302	Click to add remark
Conversion	Ru/MT	9915.321	10387.618	10755.018	Click to add remark
Variable	Ru/MT	21973.771	35095.64	23574.63	Click to add remark
Contribution	Ru/MT	31494.358	18372.489	30110.666	Click to add remark

Figure 17 Digitized AOP screen for Annual AOP Cost

AOP Approval Flow													
MEG / C2 / C2 MEG													
1 2 3 4 5 Submit Plant AOP Validate Plant AOP Review Plant AOP Approve AOP Final Approval Q2C AOP													
Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD) Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report Annual Production Plan Plant Contribution													
Plant Production Summary (T-14)													
SL.No	Item	Unit	2024-25		2025-26		Variance wrt current year budget			Variance wrt current year actuals			Remarks
	Production Volume		Budget	Actual	Budget	MT	%		MT	%			
1	EOE	MT	798076.256	788084.692	778615.5	-19460.756	-2.438		-9469.192	-1.202		-1.202	Click to add remark
2	MEG	MT	1015153	1002567.853	990398.916	-24754.084	-2.438		-12168.937	-1.214		-1.214	Click to add remark
4	Production	MT	798076.256	788084.692	778615.5	-19460.756	-2.438		-9469.192	-1.202		-1.202	Click to add remark
5	Operating Hours	Hrs	8760	8760	8520	-240	-2.74		-240	-2.74		-2.74	Click to add remark
6	ProductionRate	TPH	91.105	89.964	92.5	1.395	1.532		2.536	2.819		2.819	Click to add remark
7	NSR	Ru/MT	53468.129	53468.129	53468.129	0	0		0	0		0	Click to add remark
8	Raw Material	Ru/MT	26251.413	26386.352	26533.886	-282.473	-1.076		-147.534	-0.559		-0.559	Click to add remark
9	By Product	Ru/MT	4323.533	4307.18	4308.584	-14.949	-0.346		1.404	0.033		0.033	Click to add remark
10	RM Net of By Product	Ru/MT	21927.88	22079.172	22225.302	-297.422	-1.356		-146.13	-0.662		-0.662	Click to add remark
11	Cat-Chem	Ru/MT	45.891	348.204	1132.161	-1086.27	-2367.063		-783.957	-225.143		-225.143	Click to add remark
12	Utility	Ru/MT	9869.43	10039.414	9622.857	246.573	2.498		416.557	4.149		4.149	Click to add remark
13	Conversion	Ru/MT	9915.321	10387.618	10755.018	-839.697	-8.469		-367.4	-3.537		-3.537	Click to add remark
14	Variable	Ru/MT	21973.771	35095.64	23357.463	-1383.692	-6.297		11738.178	33.446		33.446	Click to add remark
15	Contribution	Ru/MT	31494.358	18372.489	30110.666	-1383.692	-4.393		11738.178	63.89		63.89	Click to add remark
16	Contribution Rs Lacs	Rs lacs	251348.995	144790.773	234446.316	-16902.679	-6.725		89655.543	61.921		61.921	Click to add remark

Figure 18 Digitized AOP screen for Production Plant Summary

AOP Approval Flow

Year:2025-26Vertical:MEGSite:C2Plant:C2 MEG

MEG / C2 / C2 MEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - MIS BPC (Last Budget Year), Actual Values - MIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Production (T-16)

Save

SL.No	Month	2024-25		Operating Hours		Throughput, TPH		2025-26		EO Throughput, TF	EOE Throughput, 1	Total EOE, MT	Remarks
		Production, MT											
		Budget	Actual	Budget	Actual	Budget	Actual	Operating Hours	Throughput, TPH				
1	April	63716	63142	720	720	88.49	87.7	462	116.71		91.75	42388.5	Click to add remark
2	May	68217	68410	744	744	91.69	91.95	744	116.71		91.75	68262	Click to add remark
3	June	65981	61706	720	720	91.64	85.7	720	116.71		91.75	66060	Click to add remark
4	July	68143	59922	744	744	91.59	80.54	744	116.71		91.75	68262	Click to add remark
5	August	68106	65493	744	744	91.54	88.03	744	116.71		91.75	68262	Click to add remark
6	September	65873	62067	720	720	91.49	86.2	720	116.71		91.75	66060	Click to add remark
7	October	68031	68289	744	744	91.44	91.79	744	116.71		91.75	68262	Click to add remark
8	November	65793	66493	720	720	91.38	92.35	720	116.71		91.75	66060	Click to add remark
9	December	67950	70015	744	744	91.33	94.11	744	119.57		94	69936	Click to add remark
10	January	67912	69845	744	744	91.28	93.88	658	119.57		94	61852	Click to add remark
11	February	61307	63061	672	672	91.23	93.84	672	119.57		94	63168	Click to add remark
12	March	67047	69643	744	744	90.12	93.61	744	119.57		94	69936	Click to add remark
13	Total	798076	788086	8760	8760	91.1	89.96	8416	117.67		92.5	778508.5	Click to add remark

Main Products - Production for the budget year

Products	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
EOE	42475.5	68262	66060	68262	68262	66060	68262	66060	69936	61872	63168	69936	778615.5	
MEG	54028.84	86829.26	84028.32	86829.26	86829.26	84028.32	86829.26	84028.32	88958.59	78701.18	80349.7	88958.59	990398.92	

Figure 19 Digitized AOP screen for Month Wise Production Plan

AOP Approval Flow

Year: 2025-26

Vertical: MEG

Site: C2

Plant: C2 MEG

MEG / C2 / C2 MEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval OGC AOP

* Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Monthwise Consumption (T-18)

Parameters	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	
Budgeted Selectivity for last finan...	88.15	88.06	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4	▲
Actual Selectivity for last financial ...	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4	
Guaranteed Selectivity for last fina...	86.2	85.9	85.6	85.3	85	84.7	84.3	84	83.9	83.7	83.5	83.3	
Predicted Selectivity for budget ye...	88.15	88.13	88.03	87.97	87.91	87.85	87.79	87.73	87.68	87.62	87.56	87.5	▼

RawMaterial

Particulars	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
Ethylene	MT		0.731	30979.09391	49695.03908	48173.28524	49807.31301	49849.55626	48324.72184	49917.4005	48349.90854	51254.97612	45216.73695	46209.14554	51148.03349	568925.2105
OXYGEN	MT		0.667	28003.59891	45062.29647	43800.97268	45375.3931	45493.10614	44149.26082	45725.19626	44364.95927	47262.24316	41137.92806	42300.07219	46792.44319	519467.47025

ByProducts

Particulars	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
Glycol Residue WIP	MT		0.02	906.42717	1461.59659	1409.7204	1456.71108	1456.71108	1409.7204	1456.71108	1409.7204	1492.43424	1356.94039	1348.00512	439.89744	15604.60139
MIXED PETROLE...	MT		0.014	692.05502	834.73026	872.68431	888.72277	909.10376	1000.88761	909.63962	904.17247	912.16546	982.40581	894.53405	989.09506	10790.28619
DI ETHYLENE GLY...	MT		0.102	4330.37723	6975.23984	6734.817	6959.3109	6959.3109	6734.817	6959.3109	6734.817	7068.00001	6344.32023	6384.00001	7129.9752	79314.29622

CatChem

Particulars	UOM	Spec	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total	
CORROSION INH...	kg		4.926	0	0	0	0	0	0	0	0	3835200.00021	0	0	3835200.00021	▲
ANTIFOAM UCO...	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	
CARBOHYDRAZI...	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	
CDE	kg		1.205	46902.59989	86010.12	79272	81914.4	85327.5	82575	85327.5	82575	90916.8	44488.31975	82118.4	90916.8	938344.43964
CHEM BORICACID	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	
CHEM CAT ACTIV...	MT		0	0	0	0	0	0	0	0	0	0	0	0	0	
CHEM MAXTRE...	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	
CHEM MORPHO...	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	
CHEM POTASSIU...	kg		0	0	0	0	0	0	0	0	0	0	0	0	0	

Figure 20 Digitized AOP screen for Month Wise Raw Data

AOP Approval Flow

Year:2025-26

Vertical:MEG

Site:C2

Plant:C2 MEG

MEG / C2 / C2 MEG

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval OGC AOP

* Prices - MBS BPC (Last Budget Year), Actual Values - MBS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Turnaround Details (T-19A)

SL.No

Activities

Turnaround Period

From

To

Duration, hrs

Remark

1

Shutdown XX

01-Apr-25

11-Apr-25

24000

Click to add remark

Turnaround details for the previous years data since commissioning

SL.No

Activities

Turnaround Period

From

To

Duration, hrs

Period in Months

Remark

Action

1

Shutdown XX

01-Apr-25

11-Apr-25

240

Click to add remark

Figure 21 Digitized AOP screen for Turnaround Report

Figure 22 Digitized AOP screen for Annual Production Plan

Figure 23 Digitized AOP screen for Plant Contribution

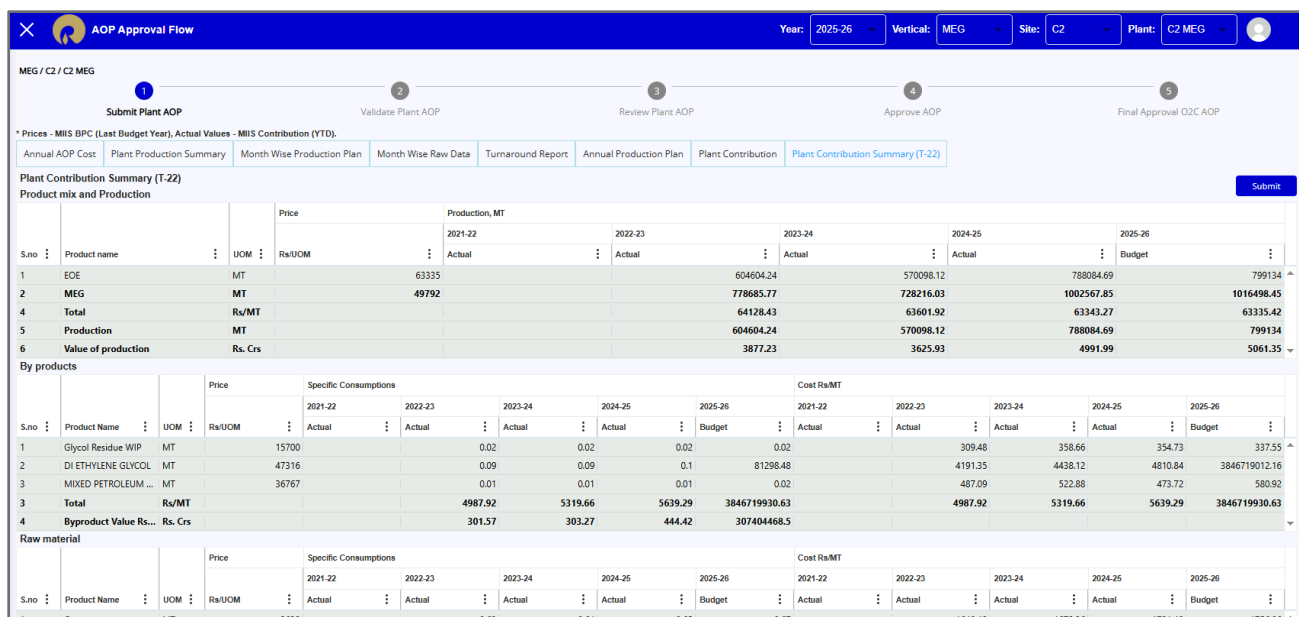


Figure 24 Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by C2 Site team